

Students' Social Media Usage Analysis

How social media use relates to sleep, academic performance, mental health and relationships

705

Students

110

Countries

4.92 h

Avg daily usage

64.3%

Say it affects academics

199

Addiction score > 7

An interactive 6-page Power BI report that analyses how students' social media usage relates to **sleep, academic performance, mental health and relationships**. Built on a survey dataset of **705 students from 110 countries**, using Power Query, a two-table data model, DAX measures, bookmarks and drill-through.

Business Problem

Different groups at an institution look at the same students through different lenses. The dashboard answers one question for each of them:

Stakeholder	What they want to know
Academic Advisors	Does social media use hurt academic performance, and for which students?
Mental Health Counselors	How do sleep, mental health and addiction scores move together?
Institutional Researchers	How do usage patterns differ by country, gender and academic level?
Students & Counselors	What does one specific student's profile look like?

Dataset

Excel workbook with two sheets, joined on `Student_ID` (705 rows each, 1:1).

Student Details — `Student_ID`, `Age`, `Gender`, `Academic_Level`, `Country`, `Sleep_Hours_Per_Night`, `Mental_Health_Score`, `Relationship_Status`, `Conflicts_Over_Social_Media`, `Addicted_Score`

Platform Details — `Student_ID`, `Avg_Daily_Usage_Hours`, `Most_Used_Platform`, `Affects_Academic_Performance`

Data quality: no nulls and no duplicate IDs in either sheet; every `Student_ID` appears in both.

Tools & Techniques

- **Power BI Desktop** — report design, bookmarks, buttons, drill-through, slicers, Edit Interactions
- **Power Query (M)** — typed columns and conditional columns
- **DAX** — measures and a calendar table
- **Data modelling** — 1:1 relationship on `Student_ID`

Data Model

Student_Details (1) — (1) Platform_Details
Student_ID Student_ID

DateTable: CALENDAR 2023-01-01 to 2023-12-31

Student_Details holds demographics and wellbeing (age, gender, country, sleep, mental health, conflicts, addiction score). **Platform_Details** holds usage behaviour (daily hours, main platform, academic impact). **DateTable** was created for future time analysis.

Calculated columns (Power Query conditional columns)

Column	Logic
Health Band	Mental_Health_Score $\geq 7 \rightarrow$ <i>Good</i> , $\geq 4 \rightarrow$ <i>Average</i> , else <i>Poor</i>
Conflict Level	Conflicts_Over_Social_Media $> 5 \rightarrow$ <i>High</i> , $> 2 \rightarrow$ <i>Medium</i> , else <i>Low</i>

DAX measures

```
Average Sleep Hours      = AVERAGE(Student_Details[Sleep_Hours_Per_Night])
Average Daily Usage Hours = AVERAGE(Platform_Details[Avg_Daily_Usage_Hours])
Total Students           = DISTINCTCOUNT(Student_Details[Student_ID])

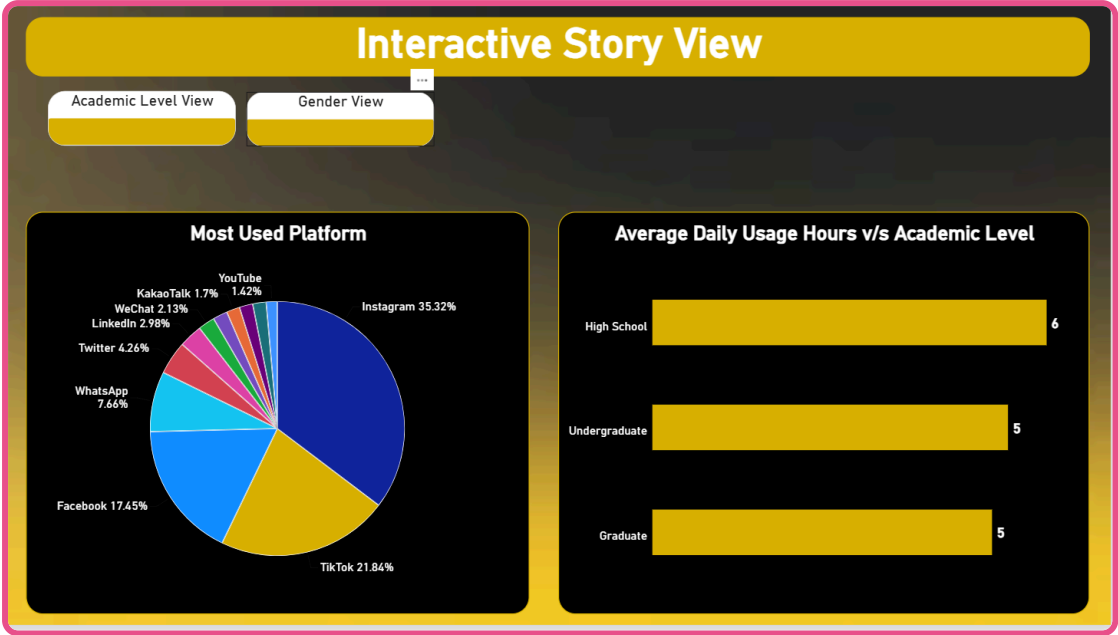
% Affected Academically =
DIVIDE(
    CALCULATE(COUNTROWS(Platform_Details),
        Platform_Details[Affects_Academic_Performance] = "Yes"),
    COUNTROWS(Platform_Details)
)

Addicted Student Count =
CALCULATE(COUNTROWS(Student_Details), Student_Details[Addicted_Score] > 7)
```

Report Pages

#	Page	Visuals
1	Executive Overview	KPI cards (Total Students, Average Usage, % Affected Academically, Avg Sleep Hours) · Avg Addiction Score by Academic Level · Avg Usage Hours by Age · Gender distribution
2	Mental Health & Lifestyle	Matrix: Country × Health Band · Scatter: Addicted Score vs Mental Health Score · Line: Avg Sleep Hours by Age · Gender slicer
3	Academic Impact	Avg Daily Usage by Academic Level · Column chart: academic impact (Yes/No) by platform · Age range slider
4	Relationships & Conflicts	Conflict Level by Relationship Status (100% stacked) · Student-wise Addiction Score table · Relationship Status distribution · Country and Academic Level slicers
5	Interactive Story View	Two bookmarked views switched by buttons — <i>Gender View</i> and <i>Academic Level View</i>

#	Page	Visuals
6	Student Profile (drill-through)	Right-click any Student_ID to see demographics, sleep, usage, addiction score, most used platform, conflicts and mental health



Page 5 — Interactive Story View (Academic Level View). Buttons switch between the Gender and Academic Level views.

Interactivity

- **Bookmarks + buttons (Page 5):** each bookmark hides one pair of charts and its slicer and shows the other pair. Buttons are kept outside both sets so they never disappear.
- **Slicer scoping:** with *Edit Interactions*, each slicer on Page 5 filters only its own platform chart, so the bar chart always shows every group for comparison.
- **Drill-through (Page 6):** filtered on `Student_ID` with a Back button.

Key Insights

- 1 The headline numbers.** Students average **4.92 hours** of daily use and **6.87 hours** of sleep. **64.3%** say social media affects their academics, and **199 students (28.2%)** have an addiction score above 7.
- 2 More screen time goes with less sleep and lower mental health.** Students using social media 3 hours a day or less average 8.2 hours of sleep and a mental health score of 7.9. Above 7 hours a day, that falls to 5.0 hours of sleep and a score of 4.9.
- 3 Addiction score and mental health score move in opposite directions** (correlation about -0.95). Students scoring above 7 on addiction average a mental health score of 5.0, against 6.7 for the rest, and sleep 5.7 hours against 7.3.

4 High school students are the most affected group. They have the highest average usage (5.54 h vs 5.00 for undergraduates and 4.78 for graduates), the highest average addiction score (8.04 vs 6.49 and 6.24) and the highest share reporting an academic impact (92.6% vs 64.9% and 61.2%). Only 27 students are in this group, so treat it as a signal rather than a firm conclusion.

5 Three platforms account for 74.5% of "most used" choices: Instagram 35.3%, TikTok 21.8% and Facebook 17.4%.

6 The platform matters for academics. Everyone whose main platform is WhatsApp (54 students) or KakaoTalk (12) reports an impact, as do 93.5% of TikTok users. LinkedIn, LINE and VKontakte users report none. Facebook (30.1%) and YouTube (30.0%) are also low.

7 Platform choice differs by gender far more than usage does. Average usage is similar (female 5.01 h, male 4.83 h), but Instagram is the main platform for 48.7% of women against 21.9% of men, and Facebook for 28.1% of men against 6.8% of women.

8 Students "in a relationship" report fewer conflicts. 47.1% of them fall in the Low conflict band, compared with 28.9% of single students and 25.0% of those in a "complicated" relationship (32 students).

Skills Demonstrated

Data loading and type checking · Data modelling (1:1 relationships) · Power Query conditional columns · DAX measures (`CALCULATE` , `DIVIDE` , `DISTINCTCOUNT`) · KPI cards, matrix, scatter, line, bar, column and pie visuals · Slicers and Edit Interactions · Bookmarks and buttons · Drill-through pages · Turning data into stakeholder-focused insights